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Paper ID: 11003

GNIOT Institute of Management Studies – PGDM (Batch 2021-23)

(TRIMESTER – I) END TERM EXAMINATION, NOVEMBER - 2021

Statistics for Business Decision

[Time: 2 Hours 30 Minutes] ***[Maximum Marks: 100]***

INSTRUCTIONS:

(i) ***There are THREE sections in this question paper.***

(ii) ***Attempt questions from each section as per the directions.***

(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A

Q.1. Attempt all parts of the following: ***(10×3=30)***

a) Sampling Vs. Population

b) What is the probability of throwing two die together the sum numbers greater than 7 with an ordinary die whose faces are numbered from 1 to 6?

c) Differentiate Probability distribution function and join Probability.

d) Define the term significance level.

e) Differentiate between a positive linear relationship and a negative linear relationship between variables

f) A Man tossed a coins four times. What is the Probability of occurrence at least three Heads?

g) State whether each of the following variables is qualitative or quantitative and indicate the measurement scales that is appropriate for each **(i)** Gender **(ii)** Class Rank **(iii)** Soft Drink Size (Small, Medium, Large)

h) Differentiate Type $\mathbf{H_0}$ **Vs** $\mathbf{H_1}$

i) Distinguish between a One-tailed and Two-tailed test.

j) Importance of Centre Limit theorem.

SECTION-B

Attempt all questions of the following: ***(4×10=40)***

Q.2.(a)'Statistics are numerical statements of facts but all facts numerically stated are not statistics. Comment upon the statement

OR

*(b)*Statistics affects everybody and touches life at many points. It is both a science and an art. Explain this statement with suitable examples.

Q.3. *(a)*The Dallas-Fort Worth Airport claims that 85% of their flights are on time. If the claim is correct, what is the probability that in a sample of 20 flights at the Dallas-Fort Worth Airport that 15 or more of the sample flights are on time'?

OR

(b) A card is drawn at random from well shuffled deck of 52 cards, find the probability that

(i) the card is either a King or diamond

(ii) A black card.

(iii) An ace or a king.

(iv) the card is not a Queen

(v) the card is either a Red card or a Black card.

Q.4.(a)(i) In which scenario you select Z score test or T student test. Also express the component of both tests.

(ii) Statistics exam scores for 20 students are as follows: 50; 53; 59; 59; 63; 63; 72; 72; 72; 72; 72; 76; 78; 81; 83; 84; 84; 84; 90; 93 Find the mode

OR

(b) Two dice are thrown simultaneously draw its Sample Space (SS). Find the probability that:

(i) Of getting sum 1

(ii) Of at least three six

(iii) Of getting a sum of 9

(iv) Of at least one 5 or sum more than 11.

(v) Of never getting sum 7

Q.5.The data on price and quantity purchased relating to a commodity for 5 months is given below:

Month :	**January**	**February**	**March**	**April**	**May**
Prices(Rs):	**10**	**10**	**11**	**12**	**12**
Quantity(Kg):	**5**	**6**	**4**	**3**	**3**

Find the correlation coefficient (r) between prices and quantity and comment on its sign and magnitude.

OR

(b) Each day, the United States Customs Service has historically intercepted about \$28 million in contraband goods being smuggled into the country, with a standard deviation of \$16 million per day. On 64 randomly chosen days in 1992, the U.S. Customs Service intercepted an average of \$30.3 million in contraband goods. Does this sample indicate (at a 5 percent level of significance) that the Customs Commissioner should be concerned that smuggling has increased above its historic level?

SECTION-C
(Numerical Question)

Q.6. Attempt all parts of the following: ***(2×15=30)***

6(a)*(a)* The height of a child increases at a rate given in the table below.

Month	1	2	3	4	5	6	7	8	9	10
Height	52.5	58.7	65	70.2	75.4	81.1	87.2	95.5	102.2	108.4

i) Find the regression equation of Month over Students height.
ii) Calculate the average increase and the standard error of estimate.

6(b) Ford company builds harvesters. For a harvester to be properly balanced when operating, a 25 lbs plate is installed on its side. The distribution of plates produced from the machine is normal. However the shop supervisor is worried that the machine is out of adjustment and is producing plates that do not average 25 lbs. To test this concern, he randomly selects 20 of the plates produced the day before and weighs them. Assume $\alpha = 0.05$. Verify if the machine is out of control. The weights of 20 plates are given in the table below.

Plate No.	Weight in lbs
1	22.6
2	22.2
3	23.2
4	27.4
5	24.5
6	27.0
7	26.6
8	28.1
9	26.9
10	24.9
11	26.2
12	25.3
13	23.1
14	24.2
15	26.1
16	25.8
17	30.4
18	28.6
19	23.5
20	23.6

Choose the appropriate value of t from the following table.

df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$
3	1.638	2.353	3.182
9	1.383	1.833	2.262
19	1.328	1.729	2.093
20	1.325	1.725	2.086